RELATIVE PSEUDO-COHERENCE FORCES QUASI-COHERENT COHOMOLOGY ABOVE THE AMPLITUDE BOUND

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 16703-16710 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 16712-16732, the complete author proof (19 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. This lemma guarantees the statement of relative pseudo-coherence actually lands in the quasi-coherent derived category; complements L16070 and L16074 on base-change behaviour.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The affine characterization of relative pseudo-coherence (same chapter, lemma-relatively-pseudo-coherence), the absolute pseudo-coherent cohomology criterion (Derived Categories of Schemes, lemma-pseudo-coherent), and the closed-immersion direct image equivalence of modules (Morphisms, lemma-i-star-equivalence; Modules, lemma-i-star-equivalence).

Source reference. Original lemma 09VC: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 16610–16622. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08KS: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 16732–16732. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01QY: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 16732–16732. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08E5: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 16732–16732. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes which is locally of finite type. If E in $D(\mathcal{O}_X)$ is m-pseudo-coherent relative to S, then $H^i(E)$ is a quasi-coherent $\mathcal{O}_X$-module for $i > m$. If E is pseudo-coherent relative to S, then E is an object of $D_{QCoh}(\mathcal{O}_X)$.*

Proof. Choose an affine open covering $S = \bigcup V_i$ and for each i an affine open covering $f^{-1}(V_i) = \bigcup U_{ij}$ such that the equivalent conditions of Lemma 09VC are satisfied for each of the pairs $(U_{ij} \to V_i, E|_{U_{ij}})$. Since being quasi-coherent is local on X, we may assume that there exists an closed immersion $i : X \to \mathbf{A}^n_S$ such that Ri_*E is m-pseudo-coherent on $\mathbf{A}^n_S$. By Derived Categories of Schemes, Lemma 08E5 this means that $H^q(Ri_*E)$ is quasi-coherent for $q > m$. Since i_* is an exact functor, we have $i_*H^q(E) = H^q(Ri_*E)$ is quasi-coherent on $\mathbf{A}^n_S$. By Morphisms, Lemma 01QY this implies that $H^q(E)$ is quasi-coherent as desired (strictly speaking it implies there exists some quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$ such that $i_*\mathcal{F} = i_*H^q(E)$ and then Modules, Lemma 08KS tells us that $\mathcal{F} \cong H^q(E)$ hence the result). □